PRE-READING:

I. Work in pairs or small groups and discuss the following:

- What is an exoplanet?
- What do you know about the term "habitable zone"?
- Why might scientists be interested in finding planets like Earth?

II. Vocabulary Match-Up

Match the scientific terms with their definitions.

Term	Definition
Exoplanet	B. A planet located outside our solar system
Spectrograph	C. An optical instrument that separates light into its component colors
Habitable zone	A. The region around a star where liquid water could exist on a planet
Biosignature	D. Evidence that might suggest the presence of life
Orbit	E. The path a planet follows around its star

III. Write five complete sentences using these terms

1. Exoplanet – There is no exoplanet yet discovered other than our Earth for ourselves to live.
2. ~~The~~ Spectograph – The Spectograph shows main three colours – Red, Green and Blue.
3. Habitable zone – Mars is not yet a habitable zone for human to colonize.
4. Biosignature – The biosignatures of various kinds of life in Earth are mesmerizing.
5. Orbit – 11.2 km/s is the speed for any object to leave the orbit of Earth.

READING:

I. Skim through the text and underline or highlight the following:

- Main ideas (major findings or points)
- Evidence, examples, data
- Names of people, instruments, and key terms

Skimming + Scanning

Researchers Confirm the Existence of an Exoplanet in the Habitable Zone

Source: The University of Oxford (News and Events)
Published: 28 Jan 2025

An international team has confirmed the discovery of a super-Earth orbiting in the habitable zone of a nearby Sun-like star. The planet was originally detected two years ago by Oxford University scientist Dr Michael Cretignier. This result, drawing on over two decades of observations, opens a window to future studies of Earth-like exoplanets that may have conditions suitable for life. The findings have been published in *Astronomy & Astrophysics*.

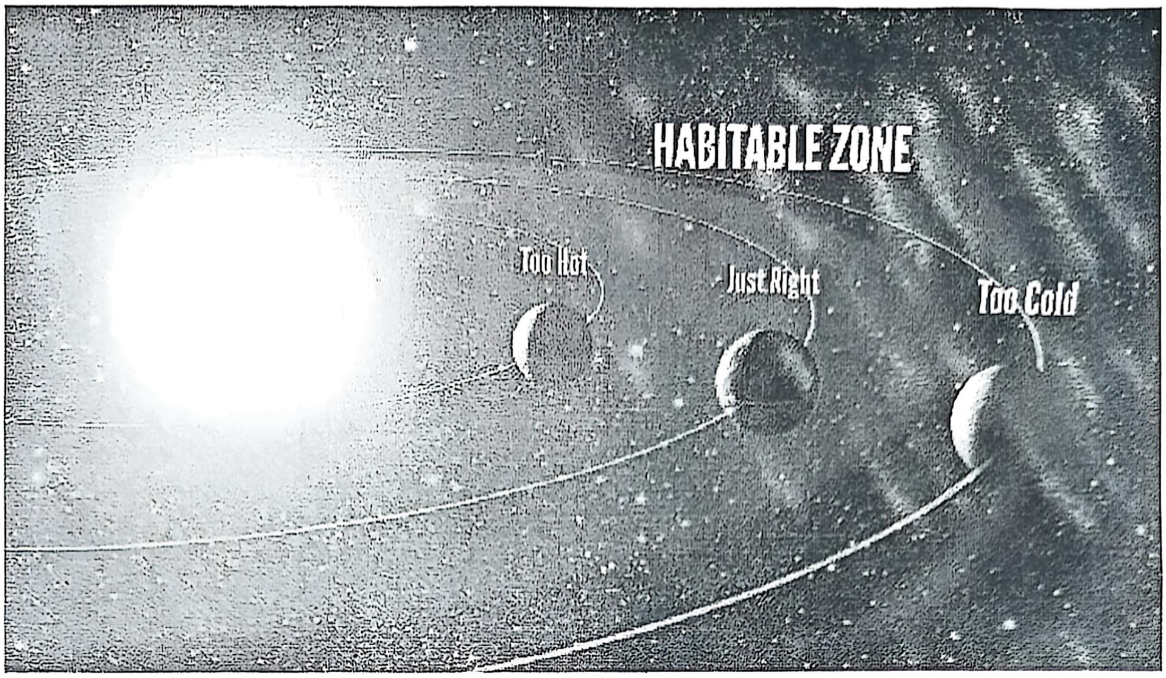
With its location in a habitable zone and relatively close proximity to Earth, this planet could play a pivotal role in future missions that will characterise the atmospheres of exoplanets to search for biosignatures indicating potential life.

The new planet, named HD 20794 d, has a mass six times that of Earth and orbits a star similar to our Sun, located just 20 light years away. Its orbit places it within the habitable zone of the system, meaning it is at the right distance from its star to sustain liquid water on its surface, a key ingredient for life as we know it.

Dr Cretignier first identified a candidate exoplanet signal in 2022, whilst analysing archived data recorded by the HARPS (High Accuracy Radial Velocity Planet Searcher) spectrograph at the La Silla Observatory in Chile. This data analyses the light absorbed and emitted by objects. Dr Cretignier spotted distinct, periodic shifts in the spectrum of light emitted by the host star, which could have been caused by the gravitational pull of a nearby planet. Due to the faintness of the signal, however, it was not clear whether this was caused by a planet, induced by the star itself, or due to an instrumental error.

To verify the signal, an international team analysed highly-precise measurements recorded over two decades by HARPS and its successor ESPRESSO (Echelle SPectrograph for Rocky Exoplanet and Stable Spectroscopic Observations), also in Chile. These instruments are among the most advanced in the world for measuring tiny variations in light spectra.

'We worked on data analysis for years, gradually analysing and eliminating all possible sources of contamination,' added Dr Cretignier. Advanced processing methods and meticulous analyses were needed to distinguish the planetary signal from background noise and subtle instrumental effects. By combining the results from the two instruments, the discovery was finally confirmed.



The orbit of HD 20794 d places it within the habitable zone of the HD 20794 system, meaning it is at the right distance from its star to sustain liquid water on its surface.

'For me, it was naturally a huge joy when we could confirm the planet's existence' Dr Cretignier said. 'It was also a relief, since the original signal was at the edge of the spectrograph's detection limit, so it was hard to be completely convinced at that time if the signal was real or not. Excitingly, its proximity with us (only 20 light-years) means there is hope for future space missions to obtain an image of it.'

Although the planet is located in the system's habitable zone, it is too early to say whether it could host life. Unlike most planets, HD 20794 d's orbit is not circular but elliptical. Its distance from its star changes significantly, causing the planet to move from the outer edge of the habitable zone to the inner edge throughout its year.

In any case, HD 20794 d will provide an invaluable test case for upcoming space projects designed to detect signs of life outside our solar system. These include the Extremely Large Telescope, Habitable Worlds Observatory, and Large Interferometer For Exoplanets (LIFE). These instruments will observe the atmosphere of nearby Earth-like planets in the habitable zone for tell-tale "biosignatures" indicative of life.

Dr Cretignier concluded: 'While my job mainly consists of finding these unknown worlds, I'm now very enthusiastic to hear what other scientists can tell us about this newly discovered planet, particularly since it is among the closest Earth-analogues we know about and given its peculiar orbit.'

The study 'Revisiting the multi-planetary system of the nearby star HD 20794. Confirmation of a low-mass planet in the habitable zone of a nearby G-dwarf' has been published in *Astronomy & Astrophysics*.

II. Work in a group/pair and complete the outline below with words/phrases by scanning the passage:

A. Overview

- Who had originally detected: Dr. Michael Creitignier
- Importance of the finding: discovery of a super-Earth.

B. Details of the Planet

- Name and type of planet: HD20794d - exoplanet
- Location and distance from Earth: 20 light years from Earth.
- Significance of orbit and water presence: liquid water on surface, elliptical

C. Discovery Process

- Instruments used: HARPS, ESPRESSO,
- Methods of analysis and verification: Spectograph detection, tiny variation in light
- Challenges in confirmation: Edge of the spectographys detectio

D. Scientific Implications

- Potential for life: Yes peculiar orbit.
- Shape and stability of orbit: Elliptical, Changes significantly.
- Relevance for future missions and studies: To gain knowledge of exoplanets.

True / False

III. Identify which of the following statements are mentioned in the passage and put a tick mark beside each:

- ✗ Scientists discovered a planet that orbits a black hole 500 light-years from Earth.
- ✓ The exoplanet HD 20794 d was confirmed after years of detailed analysis and data collection.
- ✗ Dr. Michael Cretignier built the HARPS spectrograph himself to find the exoplanet.
- ✓ The planet's orbit is elliptical, which affects its position within the habitable zone.
- ✓ The discovery was made possible by combining data from two high-precision instruments.
- ✗ The planet is covered with oceans and traces of biological lifeform is prominent.
- ✓ HD 20794 d may be useful for future missions aiming to detect life beyond Earth.
- ✗ The discovery of the exoplanet was accidental and based on a single observation.

POST READING:

I. Group Discussion:

Instructions:

Discuss the following questions in groups and prepare for a short 3-5 minutes presentation summarizing the notes of the disucussion:

- Why is HD 20794 d considered an important discovery?
- What challenges do scientists face when trying to confirm new planets?
- How does the elliptical orbit of the planet impact its potential to support life?
- Do you think we will discover life outside Earth in the near future? Why or why not?

II. Writing Prompt: "Message to the Future"

Task:

Imagine it's the year 2100, and humans have made contact with life on HD 20794 d. Write a short news article, journal entry, or letter describing:

- What was discovered
- How humans reacted
- What it means for life on Earth